How Progress Ends
by Carl Benedikt Frey

European Economic History — Spring 2026

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Roadmap

1. About the author. Is a book the right venue?
2. The central thesis
3. China's reversal of fortune
4. Europe's decentralized advantage
5. Paths to industrialization: Britain, Germany, Russia, Japan, America
6. Cold War divergence and modern stagnation
7. Critical assessment
8. Discussion

Lightning Round

In 30 seconds or less:

What did you think of the book?

About the Author: Carl Benedikt Frey

- Dieter Schwarz Associate Professor, Oxford Internet Institute
- Director of the Future of Work programme, Oxford Martin School
- Previously: *The Technology Trap* (2019). Technological change often makes life worse in the short run, while eventually benefiting everyone
- Research focus: economics of AI, technological change, and labor markets
- Ph.D. in Economic History
- Frequent contributor to policy debates on automation and inequality

Is a Book the Right Venue?

What the format enables:

- Sweeping narrative across 2,000+ years
- Bridges economics, political science, and history into one argument
- Accessible beyond academia

What it sacrifices:

- No formal model or systematic empirical test
- Evidence is illustrative, not causal
- Hard to falsify

Verdict: Yes. The scope of the argument demands a book.

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Societies that centralize too much stifle the exploration needed for breakthrough innovation. Progress can, and has, ended before.

The Framework: Exploration vs. Exploitation

Exploration (Decentralized)

- Trial and error, autonomous agents
- Many failures, occasional breakthroughs
- Markets, universities, federalism

Exploitation (Centralized)

- Scaling known solutions
- Fast catch-up, but hits a ceiling
- State planning, bureaucracies

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Key insight: Countries at the technological **frontier** need exploration; countries **behind** the frontier can grow via exploitation, but only temporarily.

Discussion

Is the boundary between exploration and exploitation this clean? Can exploitation (e.g. manufacturing improvements) itself generate innovation?

"Was Deepseek R1 an innovation in new methods of leveraging hardware, or an exploitation of algorithms developed in the US? When does the sheer scale of Chinese implementation (rolling out more solar at the beginning of 2025 than the US has in its entire history) also become its own form of innovation, rather than simply 'catching-up'?"

— Phil Bell, LSE Review of Books
https://blogs.lse.ac.uk/lsereviewofbooks/2025/10/13/
book-review-how-progress-ends-technology-innovation-and-the-fate-of-nations-carl-benedikt-frey/

Contrast: Greif, Mokyr & Tabellini, Two Paths to Prosperity

Shared ground:

- Both compare Europe and China over the long run (1000–2000)
- Both emphasize political fragmentation as key to Europe's divergence
- Both place institutions at the center, over geography (Diamond) or culture (Weber)

Where they diverge:

- **GMT:** Europe's fragmentation → warfare → fiscal-military states → capital-intensive growth. China's unity → peace → labor-intensive, Smithian growth. *Two viable paths*, not one failure.
- **Frey:** Fragmentation → innovation. Centralization → stagnation. More normatively loaded: one path ultimately wins at the frontier.
- **GMT** emphasize kinship structures: Europe's shift to universalism (weak clans, individual rights) vs. China's kinship-based organization. Largely absent in Frey.

The Needham Puzzle: Why Did China Fall Behind?

China's early lead:

- Qin unification (221 BC): centralized admin, standardized weights and script
- Printing, gunpowder, compass, paper, all centuries before Europe
- Civil service exams (605 AD): meritocratic, but focused on Confucian classics
- Song dynasty (960–1279): iron output rivaled 18th-c. Europe, paper money, textile proto-factories

What went wrong?

- Political unity killed interstate competition. No rival states to flee to
- Exam system channeled talent into bureaucracy, not commerce
- Ming sea ban (1371): Zheng He's fleet destroyed, trade criminalized
- Qing patronage: loyalty rewarded over innovation

China: The Fiscal Capacity Trap

Frey's key evidence:

- Tax revenue per capita *declined* from Song to Qing
- By 1900, China's fiscal capacity was a fraction of Europe's
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The paradox:

- Centralized enough to *block* innovation from below
- Too weak to *invest* in modernization from above

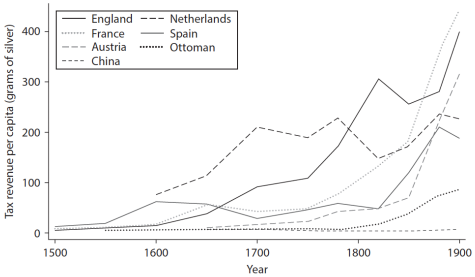

FIGURE 2.3. Comparative fiscal state capacity around the world, 1500–1900
Source: Karaman and Pamuk, "Different Paths to the Modern State in Europe."
For China: Brandt, et al., "From Divergence to Convergence," 45–123, Table 3.

Discussion

How much does Frey add to the Great Divergence debate (Pomeranz, Mokyr, Landes)?

Is the centralization lens a new contribution, or a repackaging of existing institutional arguments?

The Rise of Europe: Decentralization as Engine

Why Europe? The fall of Rome *enabled* progress.

- **Political fragmentation:** Hundreds of competing polities. Rulers who taxed too much lost productive subjects to rivals
- **Germanic traditions:** Assemblies and consensual governance. Kings needed consent, unlike Chinese emperors
- **Parliaments:** León (1188), Magna Carta (1215). Institutionalized constraints on rulers, protecting property and commerce
- **Communes:** Self-governing cities (Venice, Genoa, Hanseatic League) became hubs of trade and financial innovation
- **The Church:** Banned cousin marriage, weakening kinship networks and fostering individualism and voluntary associations

Britain: The Model of Decentralized Innovation

- Glorious Revolution (1688): Parliament controlled taxation. Credible limits on the Crown made investment safer
- Patent system (Statute of Monopolies, 1624): guaranteed inventors exclusive rights, incentivizing innovation
- Weaker guilds than on the continent. Artisans experimented freely, knowledge spread informally
- Industrial Revolution emerged *bottom-up*: Watt, Arkwright, Wedgwood were tinkerers, not state appointees
- Parliament represented merchant/industrial interests, so creative destruction was politically tolerated

Frey's key point: Britain succeeded not because the state directed innovation, but because it **could not** *easily block innovators.*

Germany/Prussia: Why Not the First Industrializer?

Why Germany lagged:

- 300+ states before 1871 with no common market or unified patent law
- Serfdom in the East limited labor mobility
- Strong guilds slowed adoption of new techniques
- Mercantilist rulers focused on state revenue, not private enterprise

What changed?

- Zollverein (1834): customs union created a large internal market
- Napoleonic reforms swept away feudal restrictions
- **Education first:** Prussia invested in mass schooling as state policy, creating a skilled workforce *ready* for industrialization

Germany/Prussia: State-Led Catch-Up and Its Limits

The catch-up model:

- State-directed industrialization: railroads, steel, chemicals
- Universal banks (Deutsche Bank) channeled savings into industry
- Legal cartels: efficient for scaling, but reduced pressure to innovate
- Humboldt universities with genuine autonomy, dominated Nobel Prizes until 1933

The catch:

- Brilliant for *replicating* British innovations, less so at the frontier
- Nazi centralization collapsed university autonomy
- Brain drain (Einstein, Fermi, von Neumann...) $\rightarrow$ US inherited the frontier

Discussion

If political fragmentation was Europe's advantage, why did economic integration (Zollverein, EU single market) not undermine innovation? Is there a tension in Frey's framework?

Britain had weak IP enforcement and rampant industrial espionage from the continent. Did that help European catch-up more than the institutional differences Frey emphasizes?

Russia and Japan: Pure Imitation, No Autonomous Research

Unlike Germany, which combined catch-up with autonomous universities, Russia and Japan had *no institutional space* for independent exploration.

Russia/USSR

- Weak state *and* weak society. Serfdom until 1861 kept most of the population uneducated
- Soviet central planning: impressive at mobilizing resources (steel output, Sputnik), but all directed from above
- Scientists worked under political control. Dissent meant exile or worse
- Result: fast catch-up, virtually zero breakthrough inventions

Japan

- Meiji state sent delegations abroad and built industries top-down
- Pilot factories privatized to zaibatsu (Mitsubishi, Mitsui), creating a tight state-business nexus
- Labor repression funded rapid capital accumulation
- Mastered incremental improvement, but few original breakthroughs

America: The Innovation Powerhouse

Why the US took the lead:

- **Federalism:** States as "laboratories of democracy" (Brandeis)
- **Constitution:** Patent clause enshrined innovation incentives from 1787
- **Democratic innovation:** Broad-based patenting, not just elites
- **Frontier individualism:** Geographic mobility, weak guilds
- **ARPA model:** Decentralized funding, high autonomy for researchers

Key contrast with Europe:

- US patent holders were far more socially diverse
- Innovation was not a preserve of the educated elite
- "Democratized" invention $\rightarrow$ more experimentation

ARPA and the Internet: Decentralized Innovation in Practice

The ARPA model (est. 1958):

- Small agency, flat hierarchy, program managers with high autonomy
- Funded high-risk research with no immediate commercial application
- Trusted researchers to define problems, not just solve assigned ones

The Internet as case study:

- ARPANET (1969): designed for resilience, not central control
- Open protocols (TCP/IP) enabled permissionless innovation
- No single firm or government directed the web's development
- Result: the most consequential innovation platform of the 20th century

Frey's strongest example: state funding + decentralized execution = breakthrough.

Cold War: The Great Experiment

USSR vs. China after 1949: Two paths of communism

Soviet Union

- Rigid central planning
- *Nomenklatura* system: political loyalty over competence
- Khrushchev's Sovnarkhoz reform failed
- Stagnation by 1970s

China (post-1978)

- Deng's *gradual* reforms
- Regional decentralization
- Special Economic Zones (Shenzhen)
- "Crossing the river by feeling the stones"

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Frey's reading: China's success was *decentralization within an autocracy* — but this has limits...

[Fig. 10.2: GDP per capita, USSR vs. China, 1920–2010]

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Autocrats **need** innovation for legitimacy and power,
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- Innovation requires free information flows — but these threaten regime control
- Autonomous researchers may produce politically inconvenient truths
- Successful entrepreneurs become independent power centers
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Historical parallel
Ming China, Nazi Germany, late USSR — all centralized at the cost of innovation.

Modern Stagnation: Not Just China

Frey argues the US faces its own innovation slowdown:

- Declining research productivity (more researchers, fewer breakthroughs)
- University bureaucratization: compliance, administration, risk aversion
- Regulatory accumulation: harder to experiment (pharma, energy, construction)
- Geographic sorting: innovation concentrated in fewer places
- Political polarization undermining institutional trust

The "veneer of progress": We may be coasting on past institutional investments while the foundations erode.

Epilogue: AI and Moravec's Paradox

Moravec's Paradox:

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Frey's provocation:

- AI is extremely good at *exploitation* — pattern recognition, optimization
- But genuine *exploration* — creative leaps, paradigm shifts — remains elusive
- AI may *accelerate* the centralization-decentralization tension
- States with AI can monitor and control more effectively...
- ...but breakthrough innovation still requires decentralized human creativity

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Will AI make the dictator's dilemma easier or harder to resolve?

Critical Assessment: Strengths

- **Ambitious scope:** 2,000+ years of history in a coherent framework
- **Clear thesis:** Centralization–decentralization as an organizing principle
- **Rich historical detail:** Moves fluently between China, Europe, US, Japan, Russia
- **Policy relevance:** Speaks directly to current debates on China, tech regulation, AI governance
- **Bridges literatures:** Connects Acemoglu & Robinson, North & Weingast, Hayek, Schumpeter, and modern growth theory

Critical Assessment: Potential Weaknesses

1. **Defining "progress":**
   Frey focuses on technological invention. But progress also includes institutions, welfare, equality, sustainability. Are these always correlated?
2. **Vague mechanism:**
   How much decentralization is optimal? The framework is qualitative and thus hard to falsify or test precisely.
3. **Geography and contingency:**
   How much is explained by institutions vs. geography, disease, demography? (Diamond, Jared; Galor, Oded)
4. **Catch-up vs. frontier distinction:**
   How does one know when to move to capitalization on existing developments? Or conversely identify the time that more decentralized exploration is needed?

Key Takeaways

1. **Progress is fragile:** History shows repeated cases of technological stagnation following centralization
2. **Catch-up $\neq$ frontier growth:** Different institutional arrangements serve different stages of development
3. **The dictator's dilemma is real:** Autocracies face an inherent tension between control and innovation
4. **Complacency is the risk:** Even democracies can drift toward institutional sclerosis

"The economic organization that excels at inventing the industrial future is not necessarily the most suitable for catching up to an existing target." (p. 13)

Thank you!

Thank you for listening!

Questions?